SynthMicro — Whole Blood Smear XAI Report

Analysis ID: 1879056f487d4cfe9eb3fc4e0252cb53
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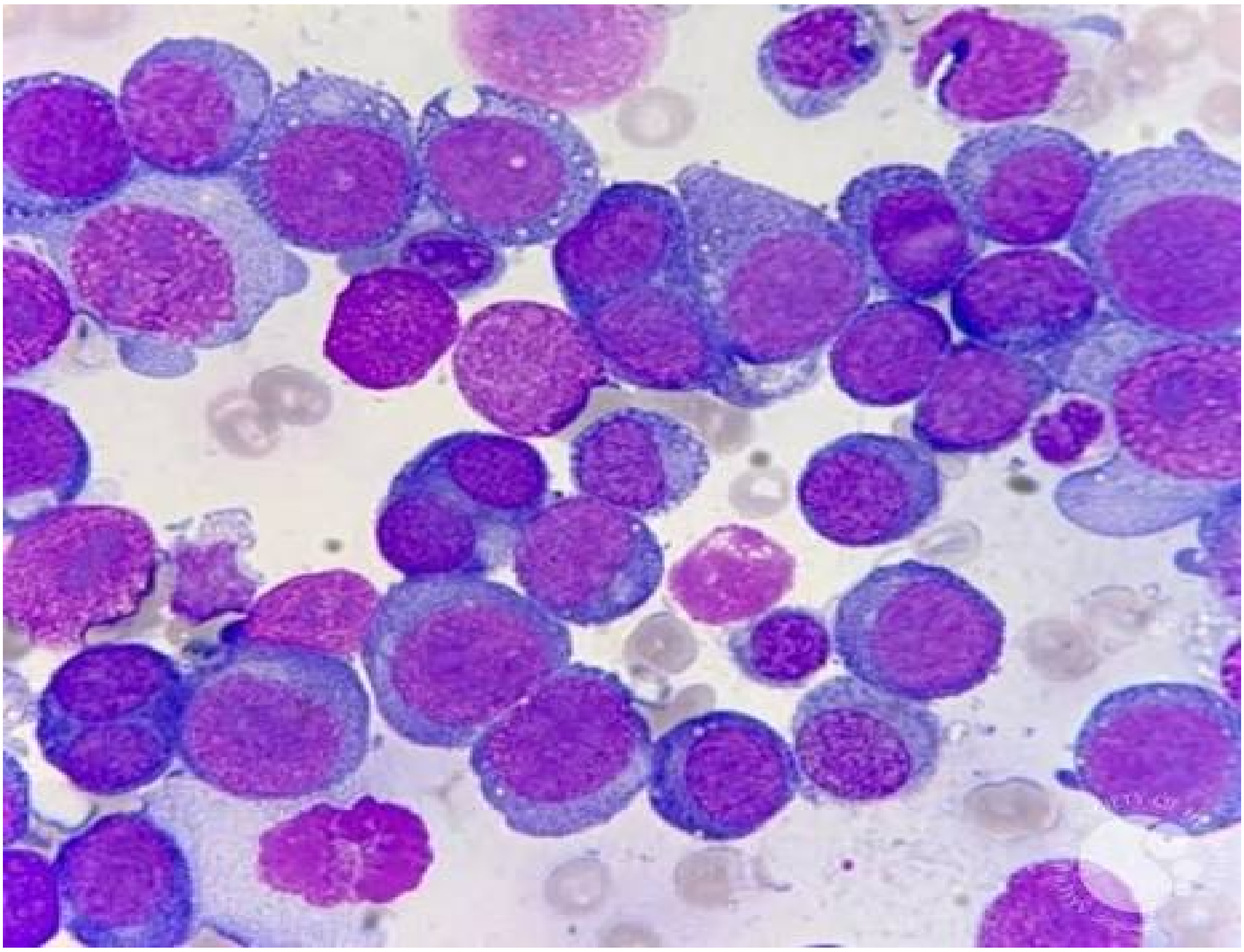
Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	48
Nucleated-cell candidates	48
Myeloblast-like candidates	42
Myeloblast-like proportion	87.50%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

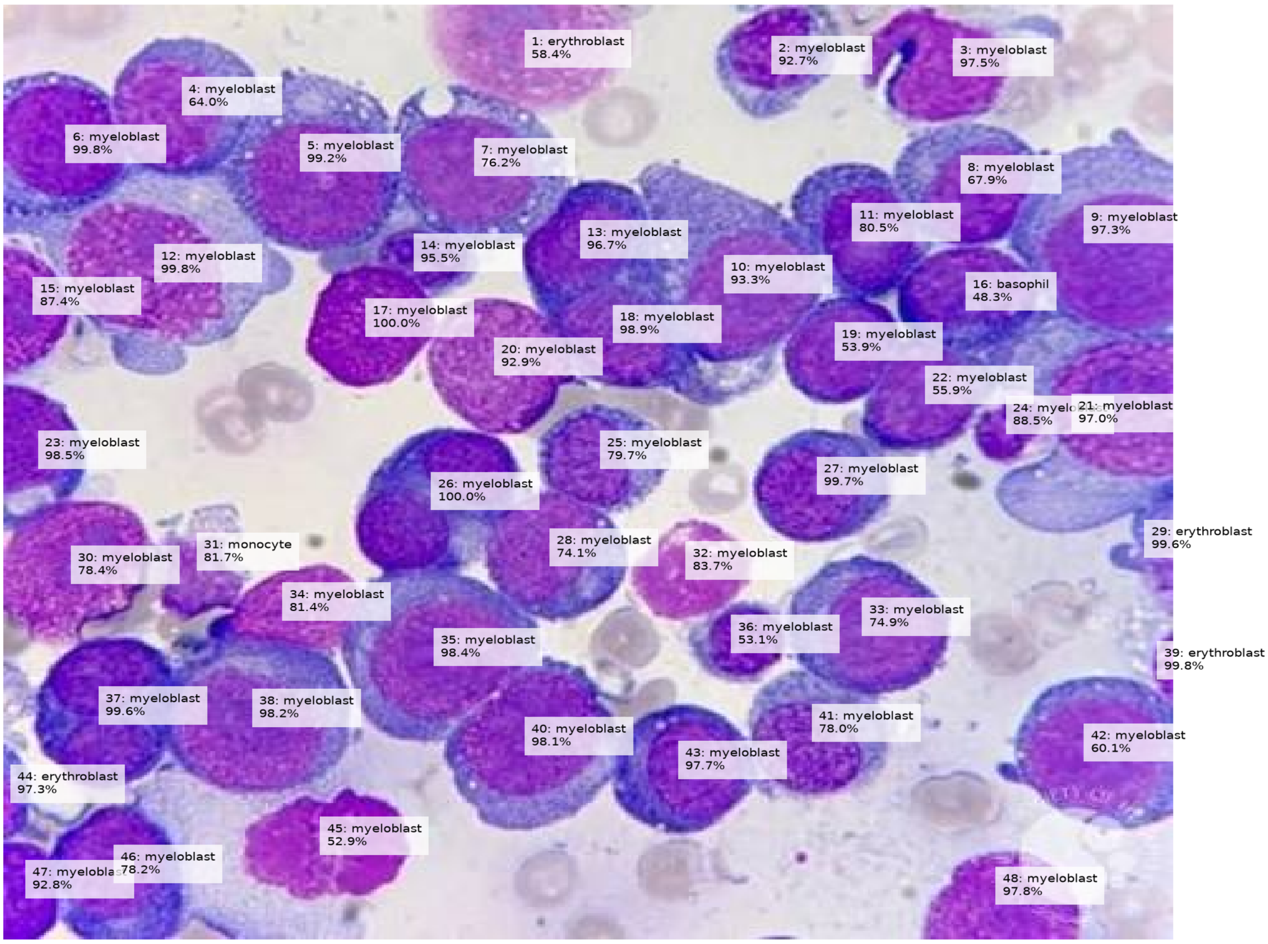
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



Cell-Level Results

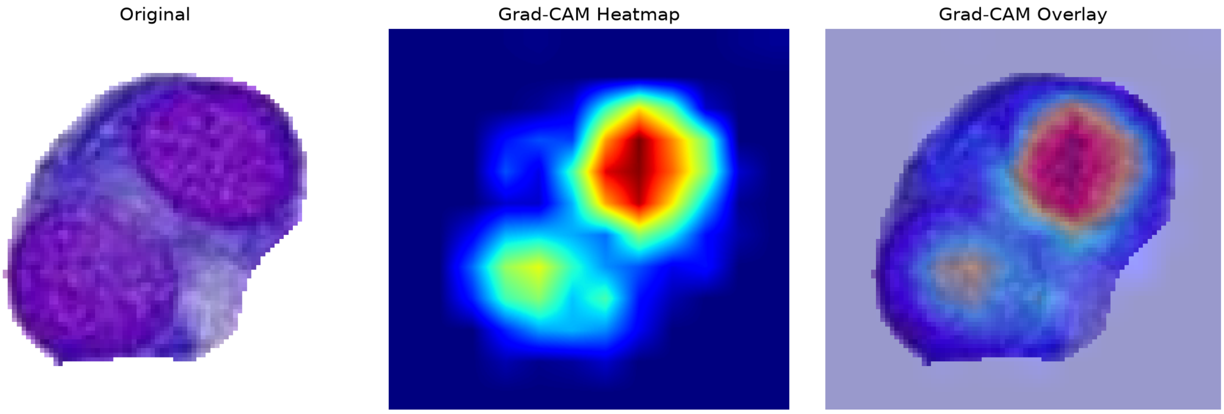
ID	Area	Purple	Class	Confidence
1	2993	68.4	erythroblast	58.4%
2	2021	73.2	myeloblast	92.7%
3	2890	90.3	myeloblast	97.5%
4	2970	84.9	myeloblast	64.0%
5	4557	81.3	myeloblast	99.2%
6	2983	111.9	myeloblast	99.8%
7	4029	70.3	myeloblast	76.2%
8	2559	85.0	myeloblast	67.9%
9	4797	83.0	myeloblast	97.3%
10	5302	67.0	myeloblast	93.3%
11	2469	90.9	myeloblast	80.5%
12	6010	70.9	myeloblast	99.8%
13	2208	94.0	myeloblast	96.7%
14	972	90.9	myeloblast	95.5%
15	1273	113.6	myeloblast	87.4%
16	2200	99.0	basophil	48.3%
17	2213	127.9	myeloblast	100.0%
18	2045	94.0	myeloblast	98.9%
19	1751	103.5	myeloblast	53.9%
20	2760	96.8	myeloblast	92.9%
21	5477	71.7	myeloblast	97.0%
22	1996	94.8	myeloblast	55.9%
23	1653	102.3	myeloblast	98.5%
24	738	98.2	myeloblast	88.5%
25	2060	75.3	myeloblast	79.7%
26	3325	97.1	myeloblast	100.0%
27	2320	86.9	myeloblast	99.7%
28	2597	84.6	myeloblast	74.1%
29	531	63.5	erythroblast	99.6%
30	3003	107.0	myeloblast	78.4%
31	1460	64.4	monocyte	81.7%
32	1645	87.8	myeloblast	83.7%
33	3236	82.0	myeloblast	74.9%
34	1580	106.0	myeloblast	81.4%
35	4525	84.8	myeloblast	98.4%
36	1139	90.1	myeloblast	53.1%
37	3146	104.8	myeloblast	99.6%
38	4329	86.6	myeloblast	98.2%
39	94	104.8	erythroblast	99.8%

ID	Area	Purple	Class	Confidence
40	3975	89.8	myeloblast	98.1%
41	2593	74.4	myeloblast	78.0%
42	3725	85.0	myeloblast	60.1%
43	2683	92.7	myeloblast	97.7%
44	331	87.9	erythroblast	97.3%
45	2589	120.8	myeloblast	52.9%
46	2734	105.5	myeloblast	78.2%
47	861	145.8	myeloblast	92.8%
48	2025	107.5	myeloblast	97.8%

Grad-CAM Explanations

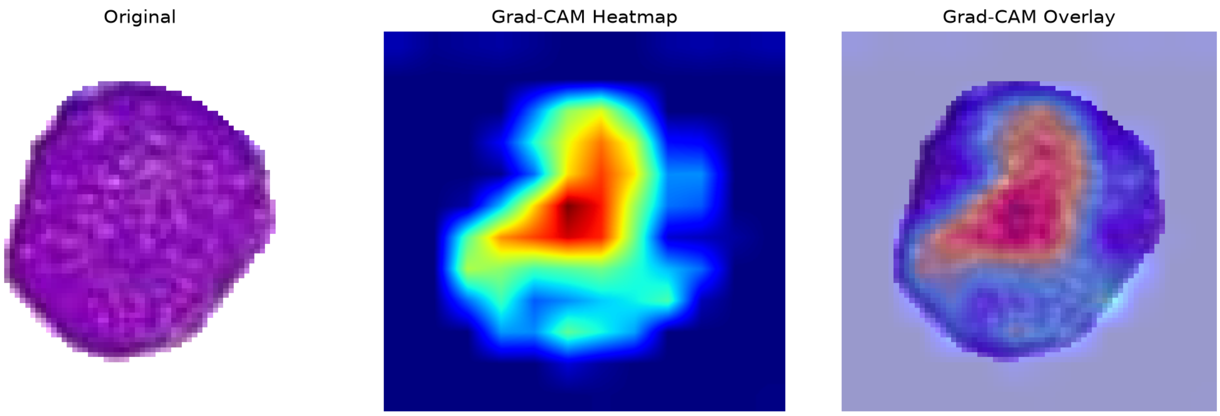
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 26 — myeloblast (100.0%)



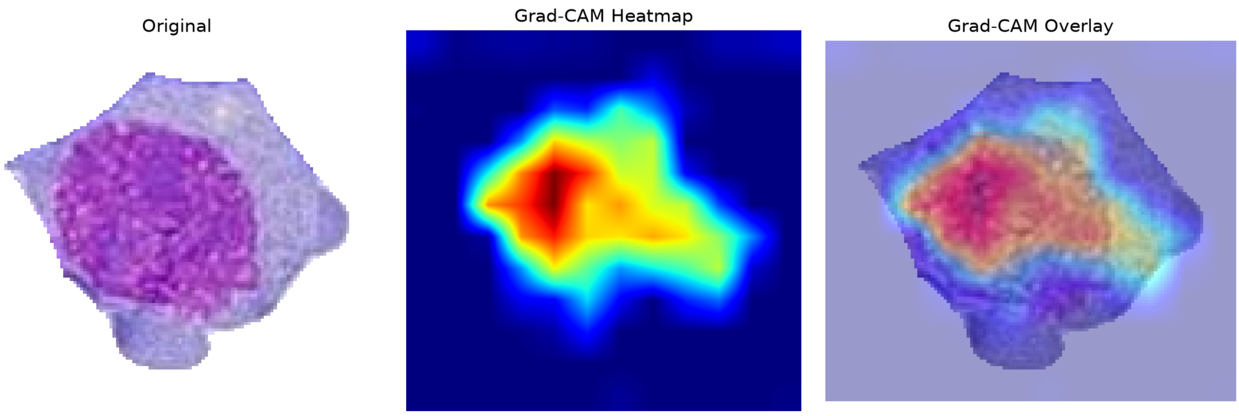
The ResNet50 model classified this segmented candidate as a myeloblast with 100.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 17 — myeloblast (100.0%)



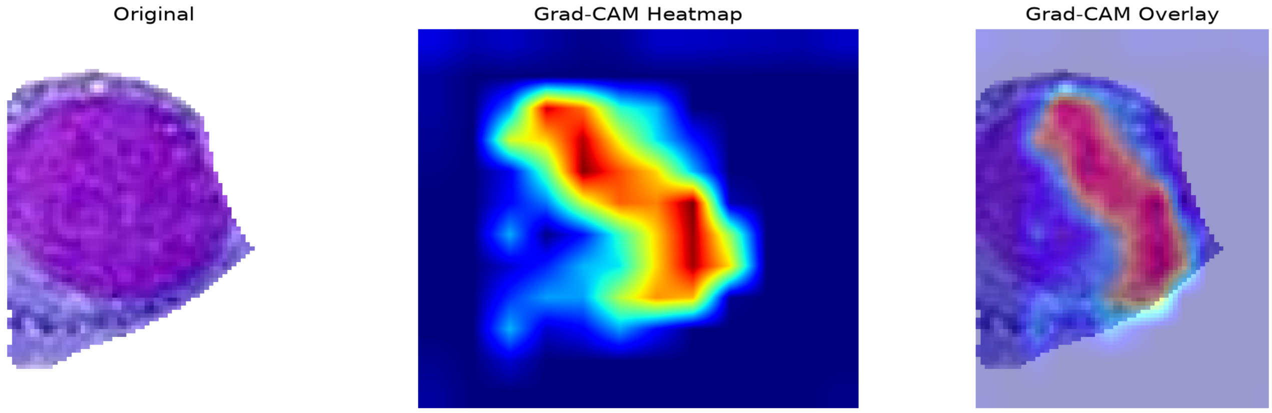
The ResNet50 model classified this segmented candidate as a myeloblast with 100.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 12 — myeloblast (99.8%)



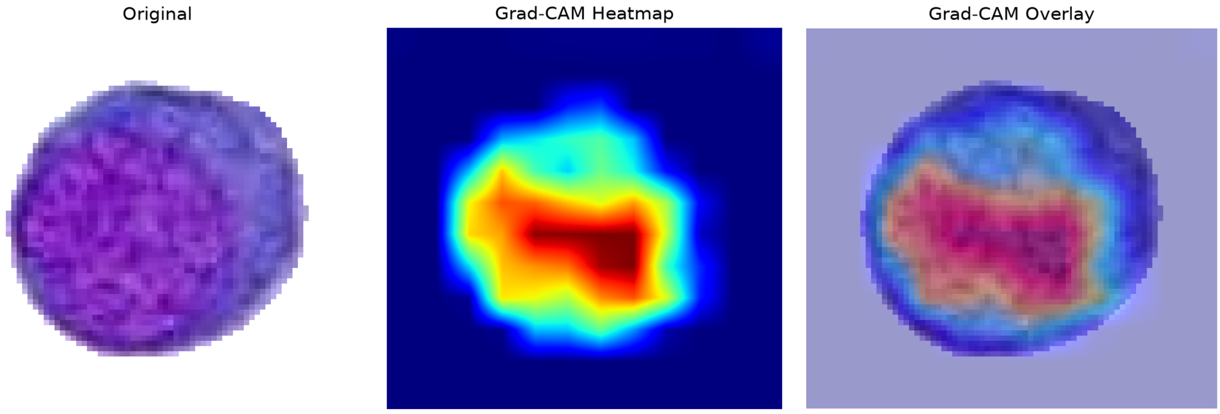
The ResNet50 model classified this segmented candidate as a myeloblast with 99.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 6 — myeloblast (99.8%)



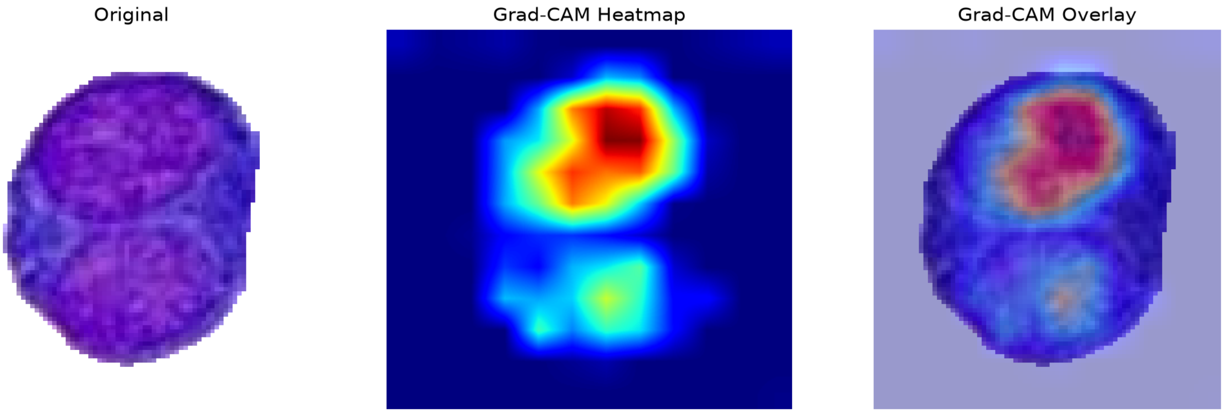
The ResNet50 model classified this segmented candidate as a myeloblast with 99.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 27 — myeloblast (99.7%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 37 — myeloblast (99.6%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.6% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

• The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.

• RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.

• Cellpose segmentation quality affects every downstream result.

• Stain, microscope, illumination and acquisition differences can cause domain shift.

• Confidence is not a calibrated probability of disease.

• A myeloblast-like prediction is not equivalent to an AML diagnosis.

• Clinical assessment requires appropriate hematopathology and laboratory testing.